JOSEPH MWANGI MAINA

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RESEARCH INTERESTS

- Trustworthy AI for health: uncertainty quantification and calibration for clinical machine learning and large language models.
- Generative models (GANs, diffusion models) for probabilistic forecasting and health data.
- Language technology for African settings, including low-resource and code-switched text.

EDUCATION

MSc, Mathematical Sciences (Data Science) *Jul 2024 – Jun 2025*
African Institute for Mathematical Sciences (AIMS) Rwanda, Kigali. Overall grade: Good Pass. Thesis: Distinction.
Thesis: comparison of conditional Wasserstein GANs with quantile regression, expectile regression and mixture density networks for probabilistic regression, scored on pinball loss, calibration and interval coverage. Data: weather and renewable energy. Supervisor: Prof. Michael Melgaard, University of Sussex. Code: github.com/jowie1111 (thesis repository).
Relevant courses: Statistical Regression, Statistical Machine Learning, Actuarial Mathematics, Numerical Analysis, Data-Driven Optimization, Introduction to Neural Networks.

BSc, Applied Statistics and Computing *Jul 2017 – Dec 2021*
Karatina University, Kenya. Second Class Honours, Upper Division.
Final-year project: factors affecting banana farming in Taveta, Kenya, using logistic regression, with flood exposure as a climatic factor.
Relevant courses: Mathematical Statistics I to III, Theory of Estimation, Introduction to Measure and Probability, Stochastic Processes, Survival Models and Analysis, Bayesian Inference and Data Analysis, Non-Parametric and Robust Methods, Statistical Computing.

PROFESSIONAL AND RESEARCH EXPERIENCE

AIMS-H2i Data Science and AI Fellow *Aug 2026 – Present*
Grow Kinesis, Lagos, Nigeria (AIMS-H2i Health Startup Fellowship). First AI/ML hire.

- Building a retrieval-augmented nutrition assistant for nutritionists, who review every suggestion before it is saved.
- Audited the production database and design files and identified data-model gaps that affect nutrition safety features.
- Built a reference food dataset from Nigerian and West African food composition tables and documented its nutrient coverage gaps.
- Wrote the `growkinesis` Python package (9 modules) for data processing.

Data Scientist Apprentice *Jan 2026 – Jul 2026*
Equity Bank Rwanda, Kigali, Rwanda

- Fine-tuned AfroXLM-R with two added classification heads to predict sentiment, aspect and intent together on Kinyarwanda, French and English customer feedback (mean F1 of 0.83), with a translation step into English so predictions could be validated.
- Forecast server demand with Random Forest and XGBoost regressors, and computed compound annual growth rates to guide procurement planning.
- Built dashboards in Power BI and Python to track issue volume, ageing and resolution.

Data Science Fellow, Transforming Healthcare Using AI *Jul 2025 – Oct 2025*
eFiche Company, Kigali, Rwanda

- Built statistical models for patient risk stratification, including Bayesian risk assessment and Cox proportional hazards models.
- Built a clinical decision-support system based on WHO IMCI guidelines for childhood illness. Led the move from a static design to a retrieval-augmented one and compared the two.

Junior Data Analyst *Dec 2023 – Dec 2024*
ICT Authority, Nairobi, Kenya

- Analysed national e-waste data to inform collection and recycling strategy, and produced reports and visuals from it.

Volunteer Data Analyst *Jun 2022 – Jan 2024*
Kenya Red Cross, Taveta, Kenya

- Wrote data-cleaning scripts for humanitarian project data and advised field teams on data collection.

PRESENTATIONS

- *Mathematical Foundations of DeepSeek-R1: Group Relative Policy Optimization (GRPO)*, MSc presentation, AIMS Rwanda, 2025. [slides]

TECHNICAL SKILLS

Programming: Python (NumPy, pandas, scikit-learn, PyTorch, TensorFlow), R (tidyverse, ggplot2, caret), SQL, Julia, LaTeX
Statistics: Regression and GLMs, quantile and expectile regression, Bayesian inference, Cox proportional hazards models, time series
Machine learning: GANs, mixture density networks, gradient boosting, random forests, neural networks, transformers, retrieval-augmented generation
Tools: Git and GitHub, Jupyter, Power BI, MongoDB

CERTIFICATIONS AND LANGUAGES

- Data Scientist Associate, DataCamp (2025). Google Data Analytics Professional Certificate, Coursera (2024).
- English (professional). Swahili (native).

REFERENCES

Available on request.